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Effect of different concentration levels of CO_2 and H_2O on the oxidation of CO: Experiments and modeling

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Abstract

Recent increased interest in oxy-fuel combustion makes it interesting to study the impact of CO_2 and H_2O concentrations on the oxidation process of CO. Experiments were performed in a quartz flow reactor operating at atmospheric pressure, over the temperature range of 700–1800 K, from fuel rich to fuel lean conditions in the presence of variable amounts of CO_2 and H_2O as representative of different exhaust gas recirculation conditions. The experimental results were simulated and interpreted in terms of a detailed kinetic mechanism by Glarborg et al. [1], with minor changes and updates. Good agreement was found between model and the present experimental results and analysis of the experimental results as well as simulation of data reported in literature on CO oxidation with CO_2/H_2O presence was supported with the model used. Sensitivity and reaction path analyses were used to identify the important reactions involving CO oxidation as function of the different reaction environments.
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Keywords: Water vapour; Carbon dioxide; Carbon monoxide oxidation; Oxy-fuel combustion; Flue gas recirculation

1. Introduction

Significant progress has been made recently in the research and demonstration of different CO_2 capture technologies. Among those, one promising option is oxy-fuel combustion, which involves the combustion of the fuel with pure oxygen and recycled flue gases instead of air as in traditional combustion. The combustion in an oxygen atmosphere, typical for oxy-fuel combustion, will result in exhaust gas compositions constituted by CO_2 and water vapour, including minor amounts of the added excess oxygen and CO, largely coming from the CO_2 to CO equilibrium, apart from the presence of minor species such as pollutants, air in-leakages and impurities in the oxygen feed. Another important difference is that the volume of flue gas is reduced by about 80% compared to air combustion due to recirculation [2]. Regarding pollutants, it has been found that oxy-fuel combustion can reduce NO_x emissions to less than about one-third of that with combustion in air [3] and also attains slightly lower SO_2 emissions [4]. The significant change in the exhaust gas composition compared to the classical air fired combustion may result in differences concerning the kinetics of oxidation as well as the formation and destruction of pollutants. Depending on the specific operating conditions, as well as on the amount of the fraction of the gas effectively recycled, the CO_2, CO, H_2O and O_2 concentrations may vary significantly.

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While the effect of different water levels for fuel oxidation has been addressed previously in a number of works (e.g. [5–7]), considerably lower work, mainly in flames [8] has been made related to the effect of different amounts of CO_2 for fuel oxidation. Recent work [9] shows experimental evidence that increased CO_2 concentration inhibits fuel oxidation, while simulation studies [10] indicate that CO_2 presence may lead to an acceleration of hydrocarbons oxidation. A review of literature data indicates a variety of results on the effect of the presence of different amounts of CO_2 and H_2O, which is usually attributed to both thermal and chemical effects of those compounds ([10] and references therein).

In this context, the present work aims to carry out a systematic experimental study in a laboratory plug flow reactor of the oxidation of CO in the presence of variable amounts of oxygen, water vapour and carbon dioxide. Experiments are performed under well-controlled conditions in a diluted atmosphere in order to assure that only chemical effects are significant. The study includes experiments in the 700–1800 K temperature range at atmospheric pressure in O_2 environments ranging from significantly rich to lean ($\lambda = 0.5$–10), and variable CO_2 (0–75%) and H_2O (0.1–10%) concentrations. Since the CO oxidation system is quite well known from a kinetic modelling point of view, the new experimental data are interpreted in terms of a kinetic model to analyze, for different stoichiometries, the presence of both low and high amounts of CO_2 and H_2O, and to evaluate the impact of the bath gas composition.

2. Experimental

The experimental installation used in the present work is described in detail elsewhere [11–15], and only a brief description is given here. A quartz plug flow reactor, according to the design of Kristensen et al. [16] is placed in a three-zone electrically heated oven, securing a uniform temperature profile throughout the reaction zone within ± 10 K. The temperature in the reaction zone was measured with a type K fine-wire thermocouple placed into a thin tube along the reactor without contact with reactants. Most of the experiments have been made in a reactor tube that has a reaction zone of 8.7 mm inside diameter and 200 mm in length. Pure gases from gas cylinders are fed to the reactor, through mass flow controllers, in up to four separate streams. CO, CO_2, O_2, and H_2O, which is fed by saturating a nitrogen stream through a bubbling water system at room temperature, and N_2 to balance up to obtain a total flow rate of 1000 (STP) ml/min. The configuration of the injection system has been designed following the investigations of Alzueta et al. [17]. At the outlet of the reaction zone, the product gas is efficiently quenched by means of external refrigeration with cooling air. The CO and CO_2 concentrations have been measured by means of an Uras14/IR analyzers. The gas stream is also analyzed by an Agilent 6890 Gas Chromatograph equipped with TCD and FID detectors. Experiments have been carried out in a different flow reactor, which is a ceramic tube in which reactants entered premixed and has dimensions of 40 mm inside diameter and 800 mm in length. This reactor has been used with success in earlier works (see for instance [18]) and allows temperatures up to 1800 K.

3. Modelling

Computations were made using both Senkin [19] from the CHEMKIN II software package [20] and CHEMKIN-PRO [21]. The reaction mechanism used is taken from the work of Glarborg et al. [1], with minor modifications and updates. The mechanism and thermodynamic data are available from the authors (uxue@unizar.es). As mentioned in the Discussion section, the results are mainly sensitive to the $H + O_2 + M$ reaction, for which the value adopted by Le Cong and Dagaut [9] has been used, and in particular to third body efficiencies chosen for CO_2 and H_2O. The third body efficiencies have been optimized in the present work according to the present experimental data and taking into account the uncertainty of this reaction rate. Another important reaction for the present results is the $CO + HO_2 \rightleftarrows CO_2 + OH$ reaction, for which the rate has been updated using the theoretical determination of You et al. [22] which is comparatively lower than the earlier value of the mechanism. The mechanism is used to analyze the experimental observations discussed in the present work. The final mechanisms with the mentioned updates and modifications has been tested with data from our group and from literature carried out under traditional air-fired conditions, and the agreement is very good.

4. Discussion of results

Flow reactor experiments of CO oxidation for different stoichiometries, CO_2 and water vapour levels have been performed, and selected literature results have been used, in order to evaluate the impact of an oxy-fuel combustion atmosphere for fuel conversion. The specific conditions of the experiments are shown in Table 1. A detailed kinetic model has been used to simulate the experimental results. In addition, sensitivity and reaction path analyses made with the model were used to identify the important reactions involving CO oxidation as function of the different reaction environments.

Table 1
Matrix of experimental conditions. All the experiments are run at a given flow rate of 1000 ml (STP)/min. Nitrogen is used to balance.

Exp.	λ	[CO] (ppm)	[O_2] (ppm)	[CO_2] (%)	[H_2O] (%)	t_r (s)	Source
1	0.5	5200	1300	0	0.1	195/T(K)	pw
2	0.5	5168	1292	25	0.1	195/T(K)	pw
3	0.5	5150	1288	0	10	195/T(K)	pw
4	0.5	5100	1275	25	10	195/T(K)	pw
5	2	5160	5160	0	0.1	195/T(K)	pw
6	2	5170	5170	25	0.1	195/T(K)	pw
7	2	5019	5019	0	10	195/T(K)	pw
8	2	5090	5090	25	10	195/T(K)	pw
9	0.7	1908	668	0	0.6	3100/T(K)	pw
10	0.7	1985	695	25	0.6	3100/T(K)	pw
11	0.7	2057	720	55	0.6	3100/T(K)	pw
12	0.7	2021	707	75	0.6	3100/T(K)	pw
13	1	2067	1034	0	0.6	3100/T(K)	pw
14	1	2020	1010	75	0.6	3100/T(K)	pw
15	2	2049	2049	0	0.6	3100/T(K)	pw
16	2	1984	1984	75	0.6	3100/T(K)	pw
17	10	2086	10,430	0	0.6	3100/T(K)	pw
18	10	2002	10,010	75	0.6	3100/T(K)	pw
19	26.7	1500	20,000	0	2	64/T(K)	[5]
20	24.6	1628	20,000	0	32.1	62/T(K)	[5]

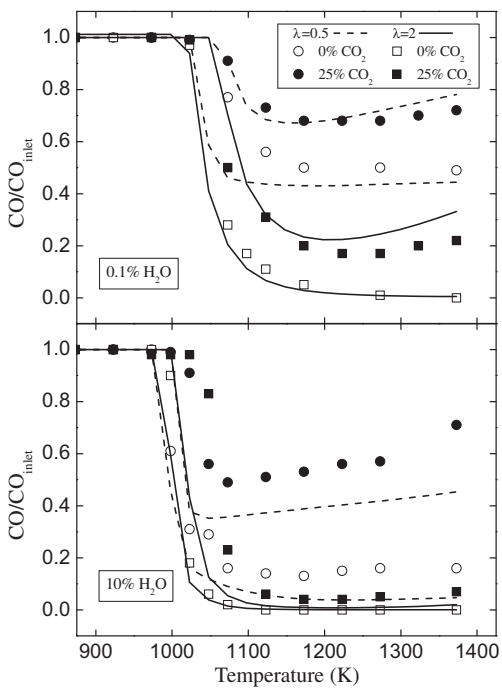

Fig. 1. Conversion of CO as a function of temperature, for different CO_2 concentrations, and for different stoichiometries, for the two considered H_2O levels. Experiments 1–8 in Table 1. Experiments: symbols; Simulations: lines.

4.1. Influence of CO_2 concentration

The influence of CO_2 concentration has been firstly considered by setting it to 0 and 25%. Figure 1 shows the conversion of CO as a function of temperature for both CO_2 levels, for two different water vapour levels, and for two different stoichiometries of 0.5 and 2. As can be observed, in general the model results match satisfactorily the experimental trends.

Oxidation of CO under classical air fired conditions is known with certain confidence (e.g. [5,23–27]), even though the presence of gases different from the classical combustion ones may affect the oxidation process in different ways, mainly by changing the O/H radical pool or acting as important collision partners in pressure dependent reactions.Under combustion conditions, CO is mainly consumed through:

$$CO + OH \rightleftarrows CO_2 + H \quad (r1)$$

and thus the concentration of OH radicals largely determines the oxidation regime of CO. In the presence of high amounts of CO_2, the reaction 1 is limited both because of equilibrium reasons and because the concentration of OH is diminished compared to the N_2 atmosphere, due to the minor extent of reaction $H + O_2 \rightleftarrows OH + O$ (r2), followed by $O + H_2O \rightleftarrows OH + OH$ (r3), partly by competition with $CO_2 + H \rightleftarrows CO + OH$ ($-$r1), reversed due to the high CO_2 levels, and being responsible for the high concentrations of CO at high temperatures.

As seen in Fig. 1, replacing nitrogen by 25% CO_2 results in a strong inhibition of CO conversion for the high H_2O level and a slight inhibition for 0.1% H_2O, for fuel rich conditions. In the absence of CO_2, the presence of high amounts of water vapour modifies significantly the composition of the radical pool being directly implied in

the fate of H radicals, $H + H_2O \rightleftarrows H_2 + OH$ (r4), thereby promoting CO oxidation through the formation of OH radicals. With 25% CO_2 the mentioned promotion is limited due to occurrence of the $CO_2 + H \rightleftarrows CO + OH$ reaction($-$r1), because of the presence of high amounts of CO_2 which are able to make reaction r1 to reverse.

Under lean conditions, the CO_2 environment produces a comparable inhibition of CO conversion for both the high and low water vapour concentrations. However at low water vapour concentration the onset temperature for CO oxidation seems not to be very affected by CO_2 presence. Under these conditions, the relatively high oxygen concentrations cause enhancement of oxidation reactions and act as a source of radicals. A new competing sequence becomes active and includes H radical consumption through the interaction with O_2 and a third body, with CO_2 as a very efficient collision partner of the reaction:

$$H + O_2 + M \rightleftarrows HO_2 + M \quad (r5)$$

Additional results of CO conversion and moderate water vapour levels are shown in Fig. 2 for both a N_2 and a 75% CO_2 atmosphere, and different stoichiometries. As seen, the model simulates very well the experimental results. It is seen that the presence of CO_2 implies an inhibition of the conversion of CO, while no significant effect is seen on the onset for CO conversion for any stoichiometry. The main effect of stoichiometry in the CO_2 atmosphere is found in the conversion of CO attained at higher temperatures, which is logically higher as the environment is leaner, while complete consumption is obtained with N_2 dilution at any stoichiometry. Also, it can be seen that for given stoichiometry and temperature values, the conversion of CO is higher in the N_2 atmosphere.

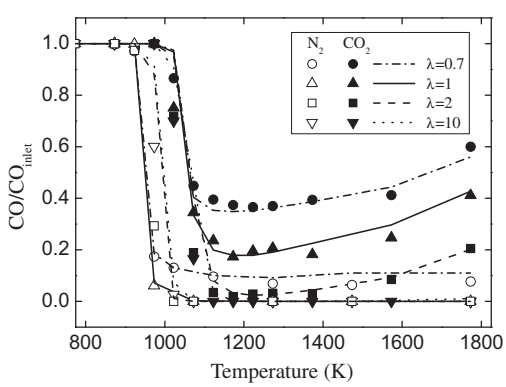

Fig. 2. Conversion of CO as a function of temperature, for different stoichiometries both in N_2 (experiments 9, 13, 15 and 17 in Table 1) and CO_2 (experiments 12, 14, 16 and 18 in Table 1). Experiments: symbols; Simulations: lines.

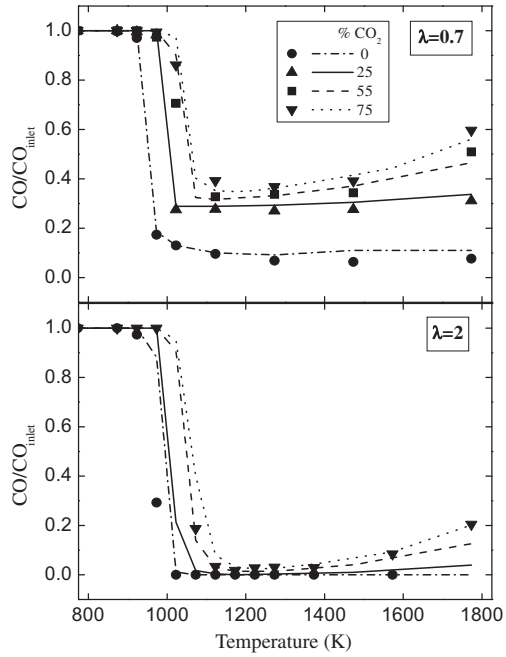

Fig. 3. Conversion of CO as a function of temperature, for two different stoichiometries and variable CO_2 concentrations (experiments 9–12, 15 and 16 in Table 1). Experiments: symbols; Simulations: lines.

The influence of different CO_2 levels on the conversion of CO for two different stoichiometries, $\lambda = 0.7$ and $\lambda = 2$, is shown in Fig. 3. An increase in CO_2 concentration from 0 to 75%, keeping the rest of conditions equal, results in an inhibition of CO conversion both in the temperature regime for reaction and in relation to the maximum CO conversion attained. This inhibition is seen to be more pronounced for lower CO_2 concentration values and for fuel-rich conditions.

4.2. Influence of water vapour concentration

The influence of water vapour concentration has been considered by changing its concentration for two different levels, 0.1 and 10%. Figure 4 shows the concentration of CO as a function of temperature for both low and high water vapour levels, respectively for a CO_2 concentration of 0% and for 25%, and for two different stoichiometries of 0.5 and 2.

In general, it is seen that the presence of high levels of water significantly promotes the oxidation of CO. For all the conditions studied, the conversion of CO is higher for the high H_2O concentration for any temperature. It is also remarkable that the observed promoting effect of water vapour on CO oxidation is less pronounced under lean conditions, due to the high radicals concentration found under these conditions. On the

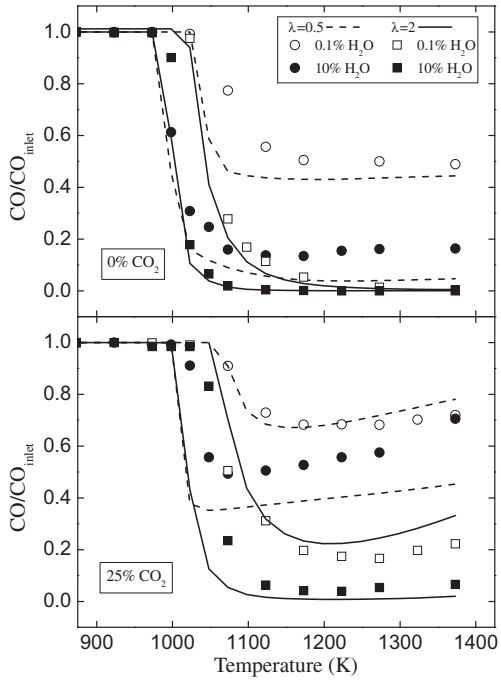

Fig. 4. Conversion of CO as a function of temperature, for different H_2O concentrations, and for different stoichiometries, for the two considered CO_2 levels. Experiments 1–8 in Table 1. Experiments: symbols; Simulations: lines.

other hand, CO oxidation is also seen to be less pronounced in the CO_2 atmosphere compared to the N_2 one, which is attributed to the important effect of CO_2 as a third body in a number of reactions of relevance.

Water vapour mainly participates in two reaction sequences: (a) directly, in the interaction with O and H radicals, through reactions 3 and 4, and by acting as a very efficient collision partner in $H + O_2 + M \rightleftarrows HO_2 + M$ (r5). R5 competes with the chain branching reaction $H + O_2 \rightleftarrows OH + O$ (r2). The competition between these two reactions largely determines the oxidation regime of CO. The efficiency of the different species that can act as collision partners in the $H + O_2 + M$ reaction is very different, with measured efficiencies for H_2O and CO_2 particularly high compared to other species. This reaction has been largely considered in literature in the past recent years (e.g. [23,28–30]), and even though huge progresses have been done, still a significant uncertainty remains. Results show that calculations are particularly sensitive to the efficiency of the third body adopted. In the present experiments, there are a number of reactions taking place simultaneously that affect significantly the CO conversion, and therefore we are not able to propose a value for the rate of $H + O_2$. However, with the selected values for the $H + O_2 + M$ reaction, we have optimized the values of the third body efficiencies for water and for CO_2, obtaining 20 and 4, respectively.

While Fig. 4 shows a promotion of CO oxidation as water vapour increases, certain literature results indicate a different behaviour. For example, in a nitrogen atmosphere, Glarborg et al. [5] show an inhibition of the CO conversion as the water vapour concentration is increased from 2 to 32%; also Le Cong and Dagaut [6] show an important inhibition of H_2 and natural gas oxidation when H_2O concentration increases from 0 to 30%. Under the very oxidizing conditions of those works, calculations show that there is an effective inhibition of fuel conversion by high levels of water vapour. As an example, Fig. 5 shows experimental results of CO oxidation by Glarborg et al. [5] and simulated results. Figure 5 additionally shows calculations made with the present model with different water vapour levels (0.1 and 10% as in the experiments carried out for the present work) keeping the rest of conditions equal as experiment 19. As seen, the model reproduces very well the experimental data and demonstrates the strong inhibition of high water vapour concentrations that is found for the significantly high O_2 levels. The temperature regime for CO oxidation changes as a function of water concentrations. The conversion of CO occurs at lower temperatures following the sequence: 2, 10, 0.1, 32.1% water vapour concentration. In general, an increase of the water concentration results in an inhibition of the CO oxidation for a given temperature. However, this trend is not always found, as proved by the results obtained at 0.1% H_2O. The inhibition of CO oxidation found, when produced, is attributed to the importance of the $H + O_2 + M$ reaction, r5, boosted by the presence of high amounts of H_2O as efficient collision

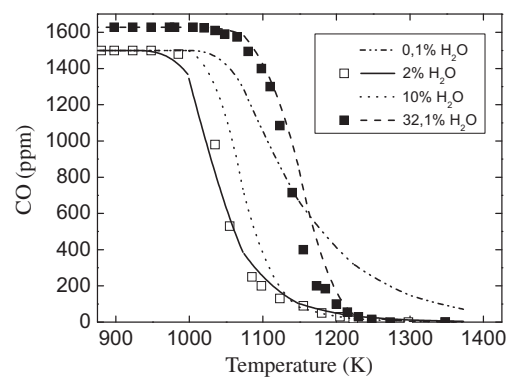

Fig. 5. Conversion of CO as a function of temperature, for different H_2O concentrations. Experimental data correspond to experiments 19 and 20 in Table 1. Experiments: symbols; Simulations: lines. Simulation of 0.1, 2 and 10% H_2O correspond to conditions of experiment 19, with the only change of water concentration.

partner. Reaction 5 produces unreactive HO_2 radicals under these conditions, that are mainly converted by interaction with OH radicals in stable species, through $HO_2 + OH \rightleftarrows H_2O + O_2$ (r6). As the water vapour concentration diminishes, reaction 5 also occurs, but N_2 has a comparative equal importance as H_2O as collision partner, as is found for the 2% concentration of water vapour. For the lowest water vapour concentration, of 0.1%, model calculations indicate that the amount of OH radicals is limited and thus the reaction $CO + O + M \rightleftarrows CO_2 + M$ (r7) appears to be competitive compared to the $CO + OH$ reaction (r1). Following the analogy of the experiments made for the present work, simulated results of Fig. 5 indicate that there is an effective promotion of CO conversion when the water vapour concentration is increased from 0.1 to 10% as found in experiments 1–8 of Table 1. This observation indicates that both results are possible and that it is the complex interaction of CO with the H/O radical pool that makes inhibiting or promoting effects to be dominant, depending on the specific conditions of temperature and mixture composition.

5. Conclusions

The experimental and modelling study of the effect of different concentration levels of CO_2 and H_2O on CO oxidation with variable amounts of O_2 has been performed under flow reactor conditions, at atmospheric pressure over the temperature range 700–1800 K. Experimental data have been interpreted in terms of detailed kinetic modelling.

Under combustion conditions the concentration of OH radicals determines the oxidation regime of CO, since it is mainly consumed through the $CO + OH \rightleftarrows CO_2 + H$ reaction. The presence of different amounts of H_2O and CO_2 in the reaction environment is found to be responsible for the modification of the O/H radical pool, boosting or suppressing CO oxidation depending on the specific operating conditions. In general the effect of carbon dioxide addition is to inhibit CO oxidation because the concentration of OH is diminished by the reverse of the reaction $CO + OH \rightleftarrows CO_2 + H$ which competes for H radicals with $H + O_2 \rightleftarrows OH + O$, and by acting as a very efficient collision partner in the $H + O_2 + M \rightleftarrows HO_2 + M$ reaction. This observed suppression is more pronounced for fuel-rich conditions and lower CO_2 concentration values. In contrast, our results show that water vapour presence enhances CO conversion, differently from some literature results with higher O_2 and H_2O levels [5,6] where an inhibiting effect was observed. Water vapour has been found to promote or inhibit CO oxidation depending on the specific conditions. Since model of CO conversion is in general quite well described by the current kinetic models, simulation of the specific conditions of any given case is suggested in order to predict the behaviour of fuel oxidation.

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